The empirical recurrence for OEIS A229592, recovered from its tail

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Abstract

OEIS A229592 records "Empirical recurrence of order 54" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 736$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 61$. Its last coefficient is $c_{54} = -262144 \neq 0$, so no further tail satisfies anything shorter; any order-54 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229592 is "Number of defective 3-colorings of a 7 X n 0..2 array connected horizontally and antidiagonally with exactly one mistake, and colors introduced in row-major 0..2 order.". It has offset 1 and begins

$$0, 26624, 839808, 25745364, 752194836, 21639043284, 613195191972, 17182547751528, \ldots$$

The entry states:

> Empirical recurrence of order 54 (see link above).

As of the "Last modified" line on the live entry (Apr 27 2021 21:29:51, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \; = \; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 736$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 736$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 54: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 54.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1472$ exact terms from $n = 7$ on, it returns order 54 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{54} c_i\, a(n-i),$$

c_1	110	c_{15}	−16613789414756	c_{29}	−147436809061851040	c_{43}	182610982696448
c_2	−5157	c_{16}	592788637848809	c_{30}	56410083150511240	c_{44}	−19333800083712
c_3	135190	c_{17}	−1701323760395806	c_{31}	79414510840124032	c_{45}	−16651957848064
c_4	−2162062	c_{18}	1492066214308157	c_{32}	−96517867521307760	c_{46}	7577290233856
c_5	20837878	c_{19}	3951612316082062	c_{33}	11132873182190176	c_{47}	−937292193792
c_6	−93686973	c_{20}	−14167359577674980	c_{34}	48269553623009872	c_{48}	−271440736256
c_7	−362215534	c_{21}	14723157123339322	c_{35}	−32624998223004448	c_{49}	141852786688
c_8	8729357816	c_{22}	13499765547358951	c_{36}	−4306757865002320	c_{50}	−30353162240
c_9	−60530565322	c_{23}	−56194088276175822	c_{37}	14859990812695936	c_{51}	3847946240
c_{10}	165212400473	c_{24}	51581154235869731	c_{38}	−5404070374710784	c_{52}	−299958272
c_{11}	561087289414	c_{25}	34990027138622320	c_{39}	−2113614938112768	c_{53}	13369344
c_{12}	−7430226974955	c_{26}	−121876173533426904	c_{40}	2304215007740288	c_{54}	−262144
c_{13}	33424155695416	c_{27}	81291829878054008	c_{41}	−425254430100992		
c_{14}	−71880836582734	c_{28}	68463460431073812	c_{42}	−300052783975680		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 61$, and it is the recurrence OEIS A229592 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{54} - c_1 x^{54-1} - \cdots - c_{54}$. Its constant term is $-c_{54} = 262144$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 54 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 54, so $\deg q = 54$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 54, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 54, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -262144.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-54 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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